Lecture 12

Cuts, Cauchy integrals and causality

Objectives. *A cut specifies the domain of a branch and separates boundary values of a Cauchy integral. We deform an integration segment to continue that integral across a cut. We then apply a boundary formula to a causal response, with the decay and regularity needed for its real and imaginary parts.*

By the end of this lecture you will be able to:

- Specify the cut and argument range of a complex power.
- Continue a Cauchy integral by a deformation that crosses no density singularity.
- Determine the sign and leading decay rate in the scalar resonance equation.
- Derive Kramers–Kronig relations under the stated response assumptions.

12.1 Branches and the quadratic root

The principal logarithm is $\log z = \ln|z| + \mathrm{i}\arg z$, with $-\pi < \arg z \le \pi$. Its holomorphic domain excludes the nonpositive real axis. A different *branch* means a holomorphic choice with a stated domain and argument range. On a chosen logarithm branch define $z^c = \exp(c\log z)$. Changing the logarithm by $2\pi\mathrm{i}n$, for $n \in \mathbb{Z}$, multiplies this power by $e^{2\pi\mathrm{i}nc}$.

For real $a < b$, consider the quadratic root

$$R(z) = \sqrt{z-a}\sqrt{z-b}, \qquad R(z)^2 = (z-a)(z-b). \tag{12.1}$$

For any function with such limits, the notation $h(x \pm \mathrm{i}0) = \lim_{\varepsilon\downarrow 0} h(x \pm \mathrm{i}\varepsilon)$ denotes its one-sided boundary values. Both factors initially use principal roots. Their product extends across $(-\infty, a)$: near a point of that interval it equals the negative holomorphic square root of $(z-a)(z-b)$ whose value there is positive before the minus sign. On either half of a small disk about the real point, the ratio of the product to that local root is continuous and takes only the values 1 and -1. Its boundary value fixes the sign, so it equals 1 throughout. Thus R is holomorphic on $\mathbb{C}\setminus[a,b]$. For large $|z|$, let $q(z)$ be the root near 1 of $1-(a+b)/z+ab/z^2$. The ratio $R(z)/(zq(z))$ is continuous and takes only the values 1 and -1 on the connected exterior of a large disk. It equals 1 on the positive real axis. Expanding q at infinity gives $R(z) = z - (a+b)/2 + O(1/z)$. On the cut its values are

$$R(x+\mathrm{i}0) = \mathrm{i}\sqrt{(x-a)(b-x)}, \qquad R(x-\mathrm{i}0) = -\mathrm{i}\sqrt{(x-a)(b-x)}, \quad a < x < b.$$

The signs follow from the two principal factors: the first is positive, and the second approaches the negative real axis from above or below. The other root is $-R$. These two choices are called its two sheets. The principal root of the whole polynomial would give a different cut. Continuation along general curves is developed in Part II.

The individual cuts and the cut of their extended product are different. The shared part of the two rays disappears from the product's cut.

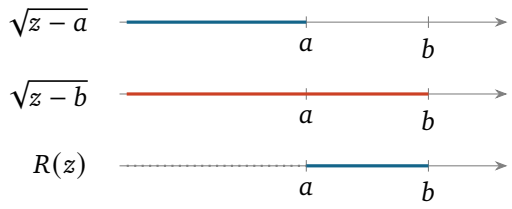


Figure 12.1: Each principal square root has a left-ray cut. On $(-\infty, a)$ both factors change sign, so their product has equal upper and lower values and extends through that ray, dotted in the bottom row. Only the segment $[a, b]$ remains as the product's cut.

12.2 Cauchy integrals off their support

Let $\rho \in L^1(\mathbb{R})$. Its *essential support* is the closed set whose complement is the largest open set on which $\rho = 0$ a.e. This largest set exists: a countable base for $\mathbb{R}$ reduces the union of all such open sets to a countable union, on which ρ is still zero a.e. The Cauchy integral convention of this course is

$$F(z) = \int_{\mathbb{R}} \frac{\rho(\lambda)}{\lambda - z}\, \mathrm{d}\lambda. \tag{12.2}$$

There is no factor $1/(2\pi\mathrm{i})$ in this definition.[1]

Proposition 12.1. *The function F is holomorphic off the essential support of ρ, and*

$$F^{(n)}(z) = n! \int_{\mathbb{R}} \frac{\rho(\lambda)}{(\lambda - z)^{n+1}}\, \mathrm{d}\lambda.$$

Proof. If the support is empty, then $\rho = 0$ a.e. and $F = 0$, so the claim is immediate. Otherwise a compact subset of the complement has positive distance c from the closed support. Choose a representative zero off that support. The integrand is bounded by $|\rho|/c$ there. The holomorphic parameter integral theorem, Theorem 10.9, applies. Its derivative formula gives the displayed expression. ∎

For example, $\rho(\lambda) = 1/(1+\lambda^2)$ gives

$$F(z) = -\frac{\pi}{z+\mathrm{i}} \quad (\operatorname{Im} z > 0), \qquad F(z) = \frac{\pi}{\mathrm{i} - z} \quad (\operatorname{Im} z < 0).$$

The residue theorem, Theorem 11.3, evaluates a positive boundary integral as $2\pi\mathrm{i}$ times its enclosed residues. To compute the upper value close the real line in the upper half-plane. Initially take $z \neq \mathrm{i}$. The residues at i and z are $1/(2\mathrm{i}(\mathrm{i} - z))$ and $1/(1+z^2)$. Their sum times $2\pi\mathrm{i}$ simplifies to the stated value. The arc integral vanishes as its radius tends to infinity. Continuity gives the value at $z = \mathrm{i}$ too. Conjugation or the lower contour gives the other value. The difference on the real axis is $2\pi\mathrm{i}/(1+\lambda^2)$.

[1] Ablowitz–Fokas use $\Phi = F/(2\pi\mathrm{i})$ and call the left of the oriented contour its plus side.

12.3 Continuation by deforming a segment

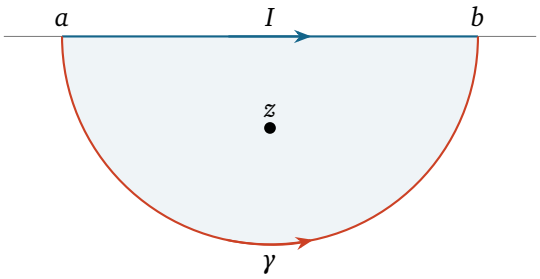
The function to be continued is the restriction of F to the upper half-plane $\mathbb{H}$. The original integral also defines values below the real axis, but these need not be the analytic continuation of that upper restriction. The rational example has already given different formulas on the two half-planes. The following construction continues the upper one through a smaller interval.

Suppose ρ agrees a.e. on a compact interval $I = [a, b]$ with a function holomorphic on a neighbourhood of the closed region between I and a lower semicircular arc γ. Denote that holomorphic extension also by ρ. Orient both I and γ from a to b. The region must contain no singularity of that holomorphic density. Define, off the remaining real support and the arc,

$$\widetilde{F}(z) = \int_{\mathbb{R}\setminus I} \frac{\rho(\lambda)}{\lambda - z}\,\mathrm{d}\lambda + \int_{\gamma} \frac{\rho(\zeta)}{\zeta - z}\,\mathrm{d}\zeta. \tag{12.3}$$

Let V be the open region between I and γ. The union $U = \mathbb{H} \cup (a, b) \cup V$ is open and connected. Define the continuation $F_{\mathrm{II}} = \widetilde{F}$ on U. This function is holomorphic through the interior of I. The first integral has positive distance from its remaining real support locally there. The second has positive distance from the compact arc. The parameter-integral argument therefore applies to each.

The diagram fixes the orientations. The boundary $\gamma - I$ is counter-clockwise around the lower region.



Both drawn curves point from left to right. The closed positive boundary uses the arc followed by the segment in reverse.

Theorem 12.2. *On $\mathbb{H}$, the continued function F_{II} equals F. Between I and γ,*

$$\widetilde{F}(z) = F(z) + 2\pi\mathrm{i}\rho(z).$$

At an interior point, $F_{\mathrm{II}}(\lambda) = F(\lambda + \mathrm{i}0)$. The original Cauchy integral F has the boundary jump

$$F(\lambda + \mathrm{i}0) - F(\lambda - \mathrm{i}0) = 2\pi\mathrm{i}\rho(\lambda), \qquad a < \lambda < b. \tag{12.4}$$

Proof. Subtract the original integral. The difference is the integral of $\rho(\zeta)/(\zeta - z)$ around $\gamma - I$. For z above the segment the kernel has no pole inside. Boundary cancellation, Lemma 10.1, makes the difference zero. For z inside the lower region remove a small disk about z. The same cancellation and the small-circle computation in Cauchy's formula give $2\pi\mathrm{i}\rho(z)$. All boundaries lie inside the open holomorphic neighbourhood of the density. The function $\widetilde{F}$ is continuous across the interior of I. The upper and lower expressions just proved therefore have limits there, and their difference is the displayed jump. ∎

For the rational example the lower expression plus $2\pi i/(1+z^2)$ simplifies to $-\pi/(z+i)$. Its pole at $-i$ prevents a deformation that sweeps across that point.

In the free quantum kernel, with mass $m > 0$ and $\hbar > 0$, the decay parameter is $\eta(z) = \sqrt{-2mz}/\hbar$, using the principal root. It has $\operatorname{Re}\eta > 0$ off $[0,\infty)$, while for energy $E > 0$,

$$\eta(E+i0) = -\frac{i\sqrt{2mE}}{\hbar}, \qquad \eta(E-i0) = \frac{i\sqrt{2mE}}{\hbar}.$$

In Theorem 8.5, this changes the decaying kernel into its two oscillatory boundary values. These signs specify the chosen sheet.

12.4 The scalar resonance equation

The Friedrichs model describes one discrete energy level coupled to a continuum of energies. Before coupling, the distinguished level has energy λ_0. The continuum is a continuous range of energies, labelled by λ. The amplitude v describes how the level couples to the continuum at each energy. Its squared modulus gives the density ρ, and the real parameter g controls the overall coupling strength. We ask where the continued scalar equation places this level after coupling. No quantum-mechanics prerequisite is needed for the scalar calculation.

For this model take $\rho = |v|^2 \in L^1(\mathbb{R})$, supported in $[0,\infty)$, with a holomorphic extension near an interval $I \subset (0,\infty)$. Let λ_0 be interior to I with $\rho(\lambda_0) > 0$. Shrink I about λ_0 so that the closed lower half-disk of the deformation lies in the holomorphic neighbourhood. The scalar continued equation is

$$\lambda_0 - z - g^2 F_{II}(z) = 0.$$

We take this scalar equation as supplied under the hypotheses above. Lecture 18 derives it only for the bounded special case in which the continuum energies lie in a compact interval of strictly positive energies, with the distinguished energy in its interior. There the reciprocal of the left-hand side is the continued scalar resolvent. Its pole describes a resonance, rather than a nonreal eigenvalue of that model's self-adjoint operator. The imaginary part is parametrised by the rate Γ below. The corresponding energy width is $\hbar\Gamma$. This scalar equation supports the golden-rule approximation cited in Quantum Mechanics II. The resonance interpretation remains sketched; its spectral derivation is not asserted here.

Proposition 12.3 (Friedrichs resonance · *sketched*). *There are $r > 0$ and $g_1 > 0$ such that, for every real $|g| < g_1$, there is exactly one zero in $\overline{D(\lambda_0, r)}$. Writing it as $E_{\mathrm{res}} - i\hbar\Gamma/2$ gives*

$$\Gamma = \frac{2\pi g^2 \rho(\lambda_0)}{\hbar} + O(g^4).$$

For small nonzero g this rate is positive.

Scalar argument. Choose a closed disk about λ_0 where F_{II} is holomorphic, and bound its modulus and derivative there by A and B. For small complex g, the map $z \mapsto \lambda_0 - g^2 F_{II}(z)$ maps this disk into itself and has Lipschitz constant at most $|g|^2 B < 1$. By Theorem 1.6, a self-map of this complete closed disk with Lipschitz constant less than one has a unique fixed point $z(g)$. Its iterates are holomorphic in g and converge locally uniformly by the geometric contraction bound. Locally uniform limits of holomorphic functions are holomorphic, as proved in § 10.6. Thus $z(g)$ is holomorphic. The bounds give

$$z(g) = \lambda_0 - g^2 F_{II}(\lambda_0) + O(g^4).$$

For real densities, $F(\lambda - \mathrm{i}0) = \overline{F(\lambda + \mathrm{i}0)}$. The jump formula therefore gives $\operatorname{Im} F_{\mathrm{II}}(\lambda_0) = \pi\rho(\lambda_0)$. Taking imaginary parts yields the stated rate. This computation locates the continued scalar zero; it does not prove an exact exponential decay law for the original self-adjoint evolution. ∎

12.5 Causality and dispersion relations

Let a response vanish for negative time. Suppose on $[0,\infty)$ that $G \in C^1$ and G, tG and G' belong to L^1. Define its frequency response by

$$\chi(\nu) = \int_0^\infty G(t)e^{\mathrm{i}\nu t}\,\mathrm{d}t. \tag{12.5}$$

The positive time-frequency sign is opposite to the spatial Fourier transform.[2] Thus $\chi(\nu) = \sqrt{2\pi}\,\widehat{\mathbb{1}_{(0,\infty)}G}(-\nu)$ on the real axis. The letter ν denotes frequency, not an integer index.

Theorem 12.4. *Under the stated assumptions on G, the response is holomorphic for $\operatorname{Im}\nu > 0$, continuous on the closed upper half-plane, and $O(1/|\nu|)$ there at infinity. For real ν its boundary values satisfy*

$$\operatorname{Re}\chi(\nu) = \frac{1}{\pi}\,\mathrm{PV}\int_{\mathbb{R}} \frac{\operatorname{Im}\chi(u)}{u-\nu}\,\mathrm{d}u,$$
$$\operatorname{Im}\chi(\nu) = -\frac{1}{\pi}\,\mathrm{PV}\int_{\mathbb{R}} \frac{\operatorname{Re}\chi(u)}{u-\nu}\,\mathrm{d}u.$$

The principal values use symmetric deletions about $u = \nu$.

Proof. In the closed upper half-plane the integrand is bounded by $|G|$. Dominated convergence (Theorem 3.5) gives continuity up to the axis. On compact subsets of the open half-plane the holomorphic parameter integral theorem gives holomorphy. The real parameter theorem, Theorem 4.4, gives the derivative on the real axis as $\int_0^\infty \mathrm{i}tG(t)e^{\mathrm{i}\nu t}\,\mathrm{d}t$, with continuity by DCT. Thus the boundary response is C^1.

Since $G' \in L^1$, the function G has a finite limit at infinity. Its integrability makes that limit zero. Integration by parts gives

$$\chi(z) = -\frac{G(0)}{\mathrm{i}z} - \frac{1}{\mathrm{i}z}\int_0^\infty G'(t)e^{\mathrm{i}zt}\,\mathrm{d}t, \qquad \operatorname{Im} z \geq 0, \quad z \neq 0.$$

In particular $|\chi(z)| \leq (|G(0)| + \|G'\|_1)/|z|$.

Fix z in the open upper half-plane and $0 < \delta < \operatorname{Im} z$. Use the half-disk centred at $\mathrm{i}\delta$ with radius $r_0 > |z - \mathrm{i}\delta|$, whose diameter lies on $\operatorname{Im}\zeta = \delta$. Remove a small disk about z. Boundary cancellation and the small-circle limit give Cauchy's formula on this shifted contour. The large arc integral tends to zero as $r_0 \to \infty$: its length is $O(r_0)$, while $|\chi(\zeta)/(\zeta - z)| = O(r_0^{-2})$ uniformly on the arc. Consequently

$$\chi(z) = \frac{1}{2\pi\mathrm{i}}\int_{\mathbb{R}} \frac{\chi(u + \mathrm{i}\delta)}{u + \mathrm{i}\delta - z}\,\mathrm{d}u.$$

Let $\delta \downarrow 0$. On bounded u intervals the denominator stays away from zero; at infinity the integrands have a common C/u^2 bound. Dominated convergence gives the same formula on the real line. Here δ is a positive shift, not the Dirac distribution.

[2] Masujima uses the opposite time-transform sign. Substituting $-\nu$ for his frequency gives the convention here.

Now put $z = \nu + \mathrm{i}\varepsilon$ and apply the Plemelj lemma, Theorem 8.2, to $\varphi(x) = \chi(\nu + x)$. This function is C^1 and its quotient by x is integrable at infinity, by the decay just proved. The plus boundary value gives

$$2\pi\mathrm{i}\chi(\nu) = \mathrm{PV}\int_{\mathbb{R}} \frac{\chi(u)}{u-\nu}\,\mathrm{d}u + \mathrm{i}\pi\chi(\nu).$$

Subtract the last term and divide by $\pi\mathrm{i}$. Real and imaginary parts give the two displayed relations. ∎

For example, $G(t) = e^{-\beta t}$ with $\operatorname{Re}\beta > 0$ meets all the assumptions and gives $\chi(\nu) = 1/(\beta - \mathrm{i}\nu)$. Its pole is $\nu = -\mathrm{i}\beta$, in the lower half-plane. Causality alone would not suffice: $G(t) = e^{t}$ on positive time does not even give the real-axis integral.

Summary.

- A branch includes its domain and argument range. The quadratic root is a product of principal factors, extended across their common cut.
- An L^1 density defines a holomorphic Cauchy integral off its support. A permitted lower deformation gives $F_{\mathrm{II}} = F + 2\pi\mathrm{i}\rho$ below the cut.
- The scalar resonance zero has a negative imaginary part at order g^2. Its spectral derivation is not asserted here; Part II develops the model.
- The assumptions $G, tG, G' \in L^1$ give the boundary regularity and upper-half-plane decay needed for the dispersion relations.

Lecture 13 uses continuation to extend Euler's Gamma integral.

References

M. J. Ablowitz and A. S. Fokas, *Complex Variables: Introduction and Applications*, second edition, 2003, §7.2, pp. 517–518. M. Masujima, *Applied Mathematical Methods in Theoretical Physics*, second edition, 2009, §§1.8 and 6.5.